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A IMPLEMENTATION OF TAPE

Algorithm 1 PyTorch snippet of TAPE.

Initialize related package

```
class TemporalAdapter(nn.Module):
    def __init__(self, merge_len, clip_num, input_dim, mid_dim, output_dim, sample_rate):
        super().__init__()
        self.AvgPool = nn.AvgPool1d(merge_len, stride = merge_len)
        self.upsample = nn.Upsample(scale_factor = sample_rate)
        self.linear_input = nn.Linear(input_dim, mid_dim)
        self.linear_output = nn.Linear(mid_dim, output_dim)
        nn.init.constant_(self.linear_output.weight, 0)
        nn.init.constant_(self.linear_output.bias, 0)
        self.Downsample_Depthwise_Separable_Conv1 = nn.Sequential(
            nn.Conv1d(mid_dim, mid_dim, merge_len*2+1, stride=sample_rate, padding=merge_len, groups=
                mid_dim),
            nn.Conv1d(mid_dim, mid_dim, 1),
            TransposeLayerNorm(mid_dim),
            nn.GELU(),
        )
        self.Downsample_Depthwise_Separable_Conv2 = nn.Sequential(
            nn.Conv1d(mid_dim, mid_dim, merge_len*2+1, stride=sample_rate, padding=merge_len, groups=
                mid_dim),
            nn.Conv1d(mid_dim, mid_dim, 1),
            TransposeLayerNorm(mid_dim),
            nn.GELU(),
        )
        self.fc = nn.Sequential(
            nn.Conv1d(mid_dim, mid_dim, clip_num+1, stride=1, padding=clip_num//2),
            TransposeLayerNorm(mid_dim),
            nn.GELU(),
        )
        self.Conv2 = nn.Sequential(
            nn.Conv1d(mid_dim, mid_dim, merge_len+1, stride=1, padding=merge_len//2, groups=mid_dim),
            nn.Conv1d(mid_dim, mid_dim, 1),
            TransposeLayerNorm(mid_dim),
            nn.GELU(),
        )
        self.Conv1 = nn.Sequential(
            nn.Conv1d(mid_dim, mid_dim, merge_len+1, stride=1, padding=merge_len//2, groups=mid_dim),
            nn.Conv1d(mid_dim, mid_dim, 1),
            TransposeLayerNorm(mid_dim),
            nn.GELU(),
        )
    def forward(self, input_tokens):
        time_ad = self.linear_input(input_tokens).transpose(1, 2)
        time_ad1 = self.AvgPool(time_ad)
        time_ad2 = self.Downsample_Depthwise_Separable_Conv1(time_ad1)
        time_ad3 = self.Downsample_Depthwise_Separable_Conv2(time_ad2)
        time_ad3 = self.fc(time_ad3)
        time_ad2 = self.upsample(time_ad3) + time_ad2
        time_ad2 = self.Conv2(time_ad2)
        time_ad1 = self.upsample(time_ad2) + time_ad1
        time_ad1 = self.Conv1(time_ad1)
        time_ad_out = self.linear_output(time_ad1.transpose(1, 2))
        return time_ad_out
```

Algorithm 1 details the implementation process of TAPE in code form. Specifically, the long video token sequence *input_tokens* is first compressed in the channel dimension by a linear layer to obtain *time_ad*, and the sequence length is compressed through a pooling layer. Next, we use a U-Net-like structure composed of one-dimensional depthwise separable convolutions to progressively down-sample the sequence, obtaining three one-dimensional temporal feature sequences with different